



KHB Technologies – Finance Operations Analytics Case Study-



Project Overview

This project is an end-to-end **Finance Operations Analytics** case study designed to solve real-world business challenges faced by KHB Technologies, a technology company operating across multiple regions and industries. The primary objective was to investigate why the company's **revenue growth was not translating into proportional cash collection**, resulting in delayed payments, higher credit risk, and inefficient working capital management.

Using **SQL, Microsoft Excel, Power Pivot, and DAX**, I transformed raw transactional data into actionable business insights by building an executive dashboard and performing detailed financial analysis across customers, products, branches, sales representatives, and payment behaviour. The project follows a consulting-style approach, focusing on solving business problems rather than simply presenting charts or writing SQL queries.



Business Objective

The finance leadership team wanted to understand why:

- Revenue continued to grow.
- Cash collection lagged behind revenue.
- Outstanding invoices kept increasing.
- Credit risk was becoming difficult to monitor.
- Collection performance varied across branches.

The objective was to identify the underlying causes and provide data-driven recommendations that would improve cash flow, strengthen credit control, and support better financial decision-making.



Tools & Technologies

- SQL

- Microsoft Excel
- Power Pivot
- DAX
- Pivot Tables
- Data Cleaning
- Financial KPI Analysis
- Dashboard Development

Project Workflow

The project followed a structured consulting methodology rather than directly jumping into data analysis.

1. Business Understanding

Identified key challenges faced by the finance department and translated them into measurable business questions.

2. Data Collection

Collected transaction, customer, payment, and product data from multiple tables.

3. Data Cleaning

Prepared the dataset by:

- Removing duplicate records
- Handling missing payment dates
- Standardizing region values
- Validating customer and product keys
- Building a calendar table
- Creating reusable DAX measures

This ensured that the analysis was accurate and reliable.

4. SQL Analysis

Performed advanced SQL analysis to answer important finance questions including:

- Which customers generate the highest revenue?
- Which customers frequently delay payments?

- Which products generate high sales but low profits?
- Which branches perform poorly in collections?
- Which invoices require immediate follow-up?
- Which customers are close to exceeding their credit limits?

5. Dashboard Development

Created an executive dashboard to monitor:

- Revenue Performance
- Collection Performance
- Customer Credit Risk
- Product Profitability
- Branch Performance
- Sales Representative Performance
- Monthly Revenue vs Cash Collection
- Invoice Ageing



Key Business Problems Solved

The analysis addressed several important business challenges:

Revenue vs Cash Collection

Although revenue was increasing, cash collections were not growing at the same pace, negatively impacting working capital.

Customer Payment Behaviour

Identified high-value customers who consistently delayed payments and contributed significantly to outstanding receivables.

Credit Risk

Detected customers who had already exceeded or were approaching their approved credit limits.

Product Profitability

Found products with high sales volumes but low profitability due to excessive discounting.

Collection Efficiency

Compared branch and sales representative performance to identify operational inefficiencies.

Invoice Prioritization

Ranked overdue invoices based on ageing to help finance teams prioritize collection efforts.



SQL Analysis Performed

- Multi-table JOINS
- Common Table Expressions (CTEs)
- Window Functions
- Aggregate Functions
- Ranking Functions
- CASE Statements
- Date Functions
- Grouping and Filtering
- Customer Segmentation
- Revenue Analysis
- Payment Delay Analysis
- Invoice Ageing Analysis
- Credit Exposure Analysis



Excel & Power Pivot Analysis

Developed interactive dashboards using:

- Pivot Tables
- Power Pivot Data Model
- DAX Measures
- KPI Cards
- Financial Metrics
- Conditional Formatting
- Interactive Filters

Key KPIs included:

- Total Revenue
- Gross Profit
- Collection Rate
- Outstanding Balance
- Bad Debt Rate
- Credit Exposure
- Monthly Cash Collection



Key Insights

The project generated several actionable business insights:

- Revenue growth alone does not guarantee healthy cash flow.
- A small group of customers contributed disproportionately to payment delays.
- Certain product lines experienced margin erosion due to excessive discounting.
- Collection performance differed significantly across branches and sales representatives.
- Some customers required revised credit policies due to increasing credit exposure.
- Revenue and cash collection trends diverged during several reporting periods, indicating inefficient working capital management.
- A prioritized invoice worklist could significantly improve collection efficiency.



Business Recommendations

Recommended improvements included:

- Prioritize collection efforts for high-value overdue customers.
- Introduce customer risk scoring for proactive credit monitoring.
- Tighten discount approval policies.
- Standardize collection practices across branches.
- Improve sales incentives by including collection performance.
- Build predictive payment delay models for future risk assessment.
- Migrate reporting to interactive business intelligence platforms.

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